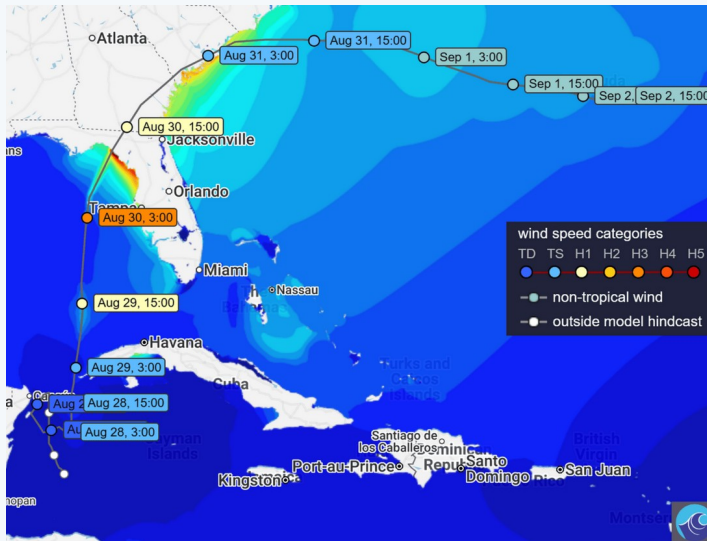


DATA

Study Area



-- Hurricane Idalia (2023)

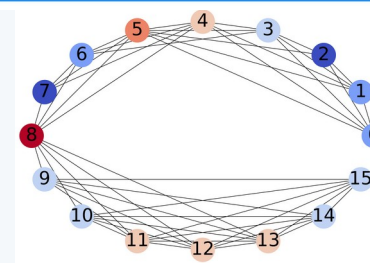
- Training: 13 historical U.S. hurricanes
- Gauge stations: NOAA, USGS, TCOON, ..
- ADCIRC model output

ANALYSES

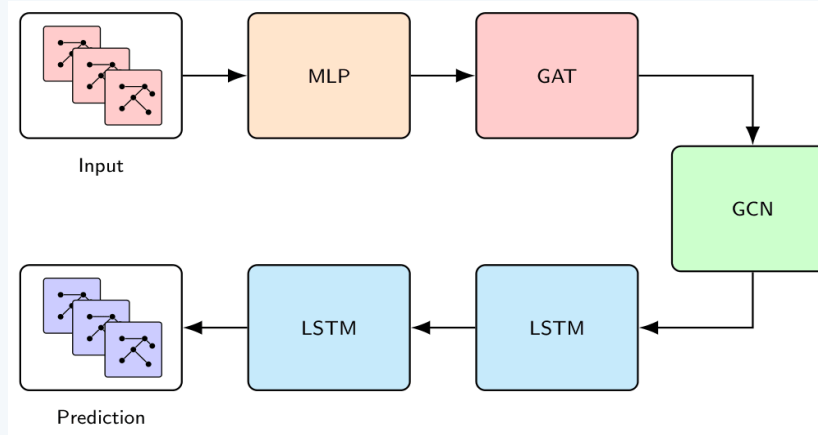
Graph Construction



- Nodes = gauge stations
- Edges = high water-level correlation + proximity

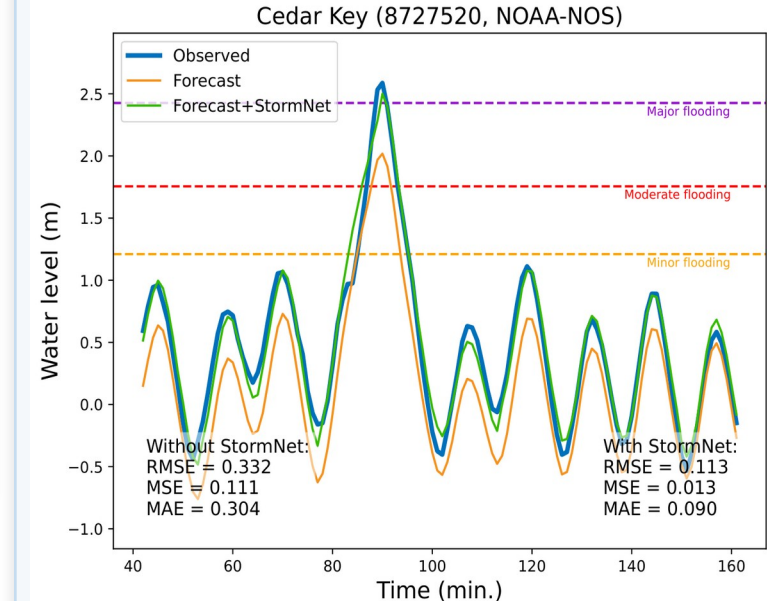


MODEL: StormNet



RESULTS

Forecast Correction



>70%

RMSE reduction
48-hr forecasts

>50%

RMSE reduction
72-hr forecasts

StormNet enables accurate, computationally efficient storm surge bias correction for real-time operational use in systems like CERA.